



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.



Capstone Project

CSA0209 - C - Programming

Banking Management System Using C Programming

Guided By:
Dr. P . ABRENA

Presented By:
N. Prince Raj (192312284)
A G. Subish (192312139)

Introduction

- The **Banking Management System** is a software application developed using the **C programming language** to automate basic banking operations.
- It enables users to perform essential banking tasks such as **account creation, deposit, withdrawal, fund transfer, and balance inquiry**.
- The system stores customer information securely using **file handling**, reducing manual paperwork and data redundancy.
- It improves the **accuracy, efficiency, and reliability** of banking transactions by minimizing human errors.
- The application provides a user-friendly interface for both customers and administrators to manage banking activities.

Objectives

- To simplify and automate daily banking operations through a computerized system.
- To provide a secure platform for managing customer accounts and financial transactions.
- To maintain accurate records of customers and transactions for easy retrieval and management.
- To minimize manual paperwork and reduce the chances of human errors in banking processes.
- To improve the speed and efficiency of banking services through automation.
- To ensure data integrity and confidentiality using proper validation and authentication.

Tools Are Used

Tool	Purpose
C Programming	Core programming language used to develop the Banking Management System
Visual Studio Code (VS Code)	Code editor used for writing and managing the source code
GCC Compiler (MinGW)	Compiles and executes the C program
HTML, CSS & JavaScript (<i>Frontend</i>)	Creates a modern, interactive graphical user interface (GUI)
File Handling	Stores customer details, account information, and transaction records
Git & GitHub	Version control and project management
XAMPP Control Panel	Local server environment used to run and manage the web application during development.
Apache	Web server that hosts the application locally and serves web pages.

Module 1: Account Management / Frontend

- Provides a user-friendly interface for easy navigation and banking operations.
- Enables customer registration and secure login authentication.
- Allows users to create and manage their bank accounts.
- Displays account details, available balance, and customer profile information.
- Provides forms for deposit, withdrawal, and fund transfer requests.
- Validates user inputs to ensure accurate and error-free transactions.
- Offers a responsive and interactive dashboard for an improved user experience.
- Provides quick access to transaction history and account activity and displays confirmation and error messages to guide users during banking operations.

INPUT (Sample Input Screen)

Create New Account

Account Number

Customer Name

Phone Number

Account Type

Initial Deposit (₹)

CREATE ACCOUNT

CLEAR

Existing Accounts (Stored in File / Database)

Acc No.	Name	Type	Balance (₹)
1001	Prince Raj	Savings	5000.00
1002	Subish A G	Current	8000.00
1003	Anitha	Savings	12000.00

PROCESSING (How the System Works)



1. User Input

Customer enters details
(Account No., Name, Amount, etc.)



2. Input Validation

Check for empty fields,
valid account number,
valid amount, etc.



3. Authentication / Login

Verify user ID and password
for secure access



4. Process Request

Perform selected operation
(Create Account / Deposit /
Withdraw / Transfer / Balance Inquiry)



5. Update File / Database

Store or update customer
and transaction records



6. Display Result

Show success message
and updated balance

OUTPUT (Sample Output Screen)

Transaction Successful



Account Number : 1001

Customer Name : Prince Raj

Transaction Type : Deposit

Amount : ₹ 2,000.00

Previous Balance : ₹ 5,000.00

Current Balance : ₹ 7,000.00

Date & Time : 22-07-2025 10:30:45 AM

Status : **SUCCESS**

Thank You for Banking With Us!

OK

Updated Account Details (Stored in File / Database)

Acc No.	Name	Type	Balance (₹)
1001	Prince Raj	Savings	7000.00
1002	Subish A G	Current	8000.00
1003	Anitha	Savings	12000.00

EXAMPLE TRANSACTION FLOW (Deposit)

1. INPUT



Account Number : 1001

Operation : Deposit

Amount : ₹ 5000



2. PROCESSING

- ✓ Account Found
- ✓ Deposit Amount Valid
- ✓ Balance Updated
- ✓ Transaction Saved



3. OUTPUT

Transaction Successful

Old Balance : ₹ 10,000

Deposited : ₹ 5,000

New Balance : ₹ 15,000



4. RECORD STORED

Transaction record
saved in file / database
for future reference.

Files Architecture

▼ BankingSystem	C transaction.c
C account.c	C transaction.h
C account.h	≡ transaction.o
≡ account.o	≡ transactions.dat
≡ accounts.dat	C utility.c
≡ bank_system.exe	C utility.h
C file.c	≡ utility.o
C file.h	
≡ file.o	
C login.c	
C login.h	
≡ login.o	
C main.c	
≡ main.o	
M Makefile	
C menu.c	
C menu.h	
≡ menu.o	
i README.md	
≡ sample_data.txt	
C transaction.c	

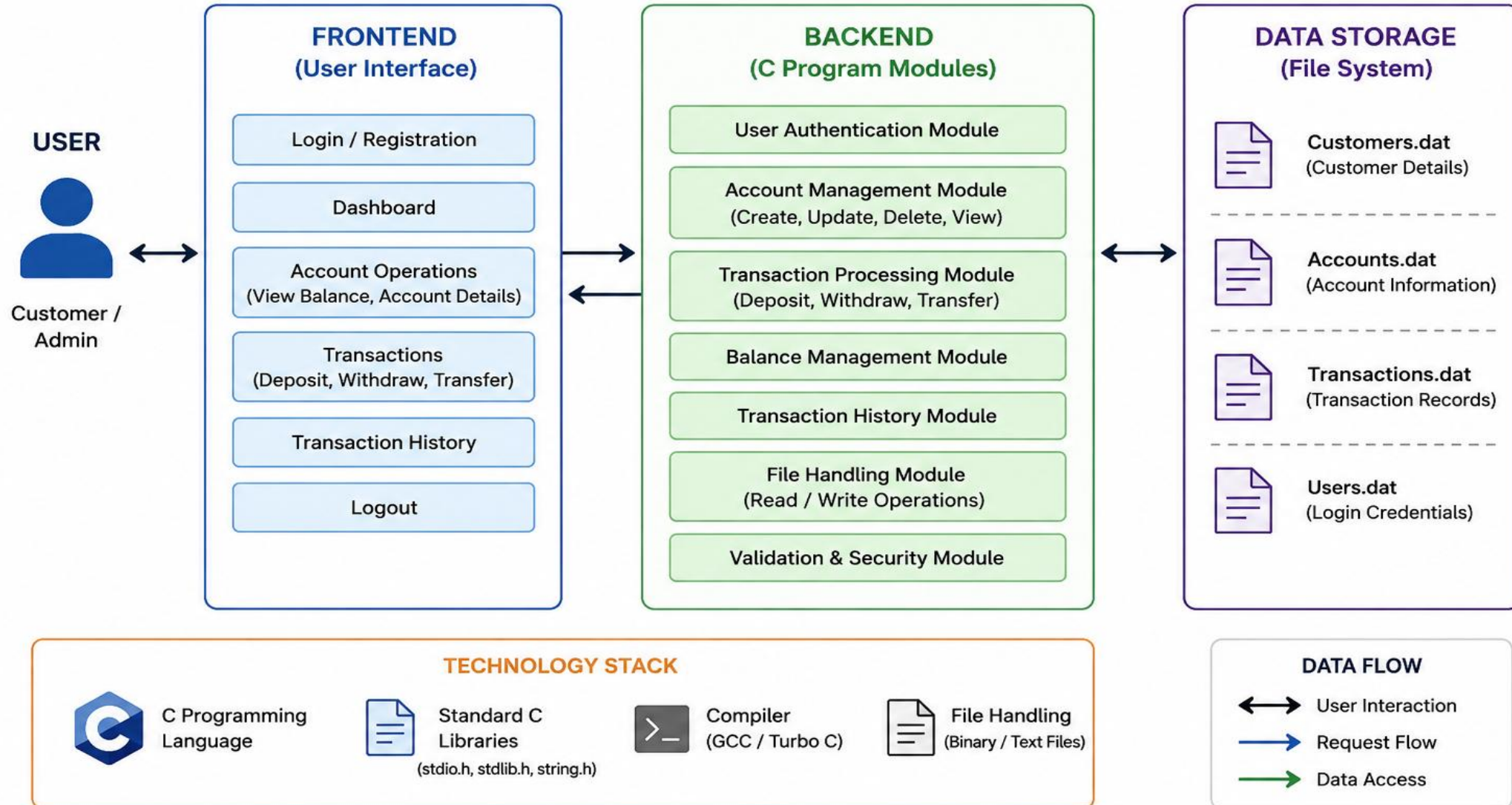
Extension	Purpose
.c	Source file containing the implementation (actual code).
.h	Header file containing declarations, macros, constants, and type definitions.
.o or .obj	Object file created after compilation.
.exe (Windows)	Executable program generated after linking.
.dat	is simply a data file that your program uses to save and retrieve information.
Makefile	Automates the compilation and build process.

Module 2: Transaction Management Module / Backend

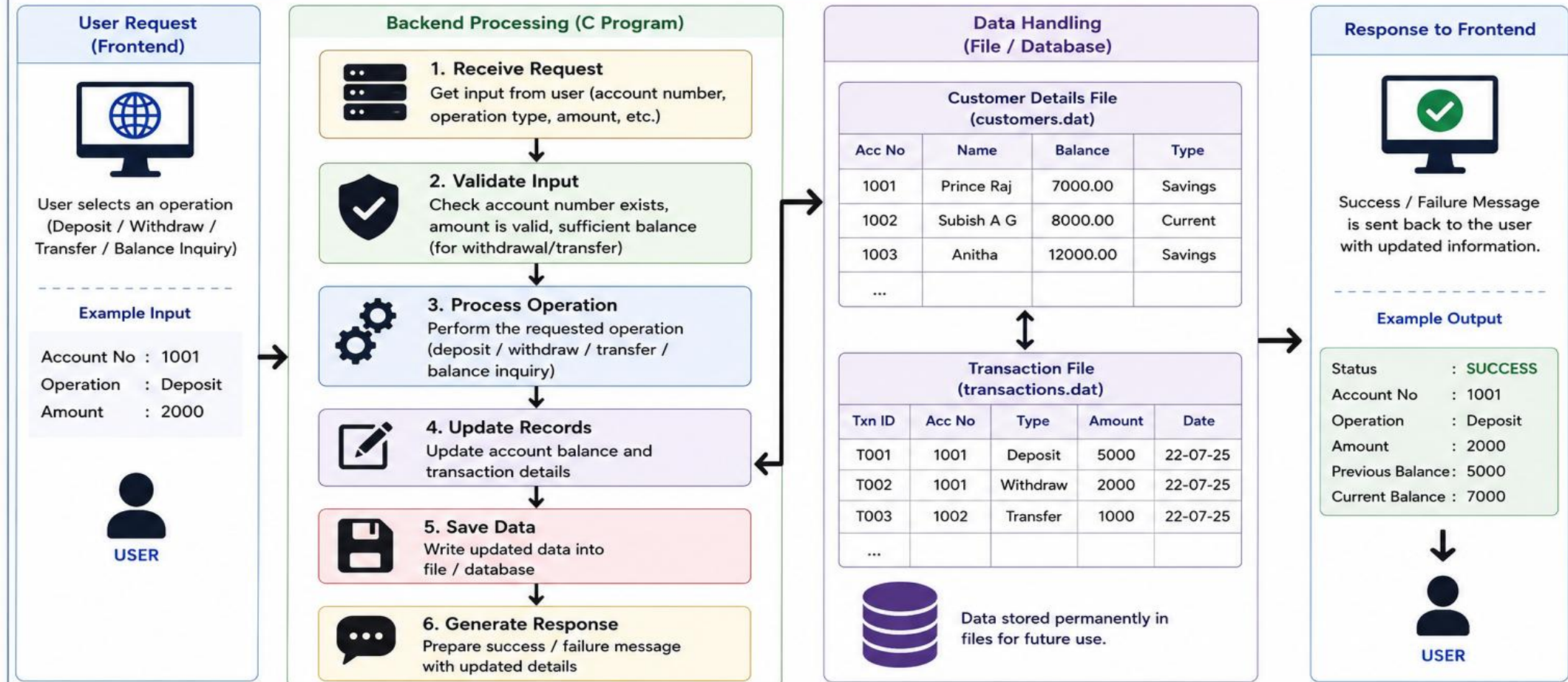
- Processes customer requests such as deposits, withdrawals, and fund transfers.
- Maintains and updates customer account balances after every transaction.
- Stores customer information and transaction records securely using file handling or a database.
- Verifies user credentials to ensure secure access to banking services.
- Validates transaction details to prevent invalid or duplicate operations.
- Generates and maintains transaction history for future reference.
- Ensures data consistency and integrity during all banking operations.

System Architecture Diagram

Banking Management System (C Programming)



BACKEND PROCESSING (TRANSACTION MANAGEMENT / SERVER SIDE)



EXAMPLE FLOW (DEPOSIT)



Key Features

- Secure user login and authentication.
- Easy customer account creation and management.
- Fast and accurate deposit, withdrawal, and fund transfer operations.
- Real-time balance inquiry and account information display.
- Automatic recording and maintenance of transaction history.
- Secure storage of customer data using file handling or a database.
- User-friendly interface for smooth and efficient banking operations.
- Error handling and input validation to ensure reliable transactions.

Outputs

```
=====
WELCOME TO THE BANKING MANAGEMENT SYSTEM
=====
Press Enter to continue to the login screen...
|
```



```
=====
ADMINISTRATOR LOGIN
=====
Username: admin
Password: admin123

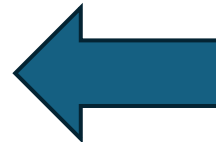
Login Successful! Welcome, admin.

Press Enter to continue...|
```



```
=====
Enter your choice (1-11): 11

Thank you for using the Bank Management System. Goodbye!
|
```



```
=====
1. Create Account
2. Display Accounts
3. Search Account
4. Deposit Money
5. Withdraw Money
6. Transfer Money
7. Balance Inquiry
8. Update Account
9. Delete Account
10. Transaction History
11. Exit
=====
Enter your choice (1-11): |
```

BANKING MANAGEMENT SYSTEM – RESULT

The system successfully performs all banking operations and gives the expected output with updated records.

1. Account Creation (Input)

Create New Account

Account Number : 1001

Customer Name : Prince Raj

Phone Number : 9876543210

Account Type : Savings

Initial Deposit (₹) : 5000

CREATE ACCOUNT CLEAR

Result : Account created successfully!

2. Login (Input)

User Login

Account Number : 1001

Password :

LOGIN CANCEL

Result : Login successful!

3. Main Menu (After Login)

Banking Management System

WELCOME, Prince Raj (1001)

Account Type : Savings

Current Balance : ₹ 5,000.00

1. Deposit 2. Withdraw 3. Fund Transfer 4. Balance Inquiry 5. Transaction History 6. Account Details 7. Change Password 8. Logout

Result : User logged in and main menu displayed.

4. Deposit (Process & Output)

Deposit Money

Account Number : 1001

Deposit Amount (₹) : 2000

SUBMIT CANCEL

Transaction Successful

Amount Deposited : ₹ 2,000.00

New Balance : ₹ 7,000.00

Result : Deposit successful, balance updated!

5. Withdraw (Process & Output)

Withdraw Money

Account Number : 1001

Withdraw Amount (₹) : 1500

SUBMIT CANCEL

Transaction Successful

Amount Withdrawn : ₹ 1,500.00

New Balance : ₹ 5,500.00

Result : Withdrawal successful, balance updated!

6. Fund Transfer (Process & Output)

Fund Transfer

From Account : 1001

To Account : 1002

Amount (₹) : 1000

TRANSFER CANCEL

Transaction Successful

Amount Transferred : ₹ 1,000.00

New Balance : ₹ 4,500.00

Result : Fund transfer successful!

7. Balance Inquiry (Output)

Balance Inquiry

Account Number : 1001

CHECK BALANCE

Account Balance

Account Number : 1001

Customer Name : Prince Raj

Current Balance : ₹ 4,500.00

Result : Balance retrieved successfully!

8. Transaction History (Output)

Transaction History					
Txn ID	Date & Time	Type	Account No.	Amount (₹)	Balance (₹)
T001	22-07-2025 09:15:30	Deposit	1001	+2000.00	7000.00
T002	22-07-2025 09:30:10	Withdraw	1001	-1500.00	5500.00
T003	22-07-2025 09:45:20	Transfer	1001	-1000.00	4500.00

Result : All transactions recorded successfully!

FINAL RESULT OF THE PROJECT



- All modules are working correctly.
- User can perform all banking operations successfully.
- Data is stored and updated in files.
- Records are accurate and can be retrieved anytime.
- The system is secure, efficient and user-friendly.

FILES CREATED (DATA STORED)



customers.dat

Stores customer details such as account number, name, phone, balance, account type, etc.



transactions.dat

Stores all transaction records like deposit, withdrawal, transfer, with date and time.

SAMPLE FINAL BALANCE (Account No: 1001)

Initial Deposit	: ₹ 5,000.00
Total Deposits	: ₹ 2,000.00
Total Withdrawals	: ₹ 1,500.00
Total Transfer (Sent)	: ₹ 1,000.00

Current Balance	: ₹ 4,500.00

Conclusion

- The Banking Management System successfully automates essential banking operations using the C programming language.
- It provides a secure and efficient way to manage customer accounts and financial transactions.
- The system minimizes manual errors while improving the speed and accuracy of banking services.
- Features such as account management, transaction processing, and record maintenance enhance overall operational efficiency.
- The modular design makes the application easy to maintain, update, and expand.
- The project demonstrates the practical application of C programming, file handling, and software development concepts in solving real-world banking problems.

Thank You



Engineer to Excel

Problem Statement

1 Time-Consuming

Manually matching students to rooms takes hours and doesn't scale as intake grows.

2 Error-Prone

Duplicate allotments, double bookings, registers are common with paper records.

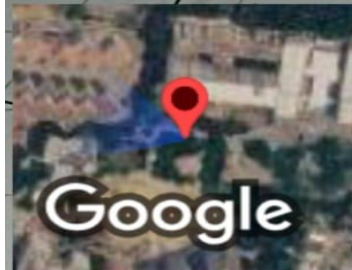
3 No Real-Time View

Wardens can't instantly see which rooms are vacant, full, or reserved.

4 Hard to Audit

Tracking who stayed where, and for how long, is difficult without digital records.

- Hostel room allocation in many colleges is still performed manually, leading to delays, duplicate assignments, and difficulty in managing room availability.
- An automated Hostel Room Allotment System is needed to efficiently manage student rooms, track vacancies, and maintain accurate hostel information while reducing errors and improving overall efficiency.



Mevalurkuppam, Tamil Nadu, India



22h8+7xq, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.02752° Long 80.01694°
Thursday, 30/07/2026 02:43 PM GMT +05:30

 GPS Map Camera